

Mira Askar

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EDUCATION

University of Warwick

BEng Electrical and Electronic Engineering

Coventry, United Kingdom

Sep 2025 – Present

Expected graduation: July 2028 | First-year average: 74.8% | Engineering Mathematics: 91%

Malvern College

A Levels

Sep 2023 – Aug 2025

Grades: A*AA

WORK EXPERIENCE

Healthware

Java Backend Developer

London, United Kingdom (Remote)

Aug – Sep 2026

- Developed Java/Spring RESTful API workflows and endpoint contracts to ingest medication selections and patient allergy histories from hospital systems via structured JSON payloads.
- Designed response DTOs and API schemas to return allergy-risk assessments, contributing factors and safer medication alternatives to downstream clinical systems.
- Defined input-validation rules, exception-handling flows, HTTP status codes and structured error responses to support reliable exchange of sensitive clinical data.

Trans Asia Construction LLP

Java Backend Developer

Kyzylorda, Kazakhstan

Jul – Aug 2026

- Developed a Java/Spring backend for managing employee records, using Hibernate ORM and PostgreSQL, with Docker for containerised deployment.
- Designed relational data models and CRUD workflows for employee records and document-status tracking, enabling structured search and administration.
- Implemented server-side validation, database queries and update workflows to improve data consistency and reduce manual file-based record management.

PROJECTS

Hyperloop University Project

Sep 2025 – Present

Electronics Hardware Engineer

- Simulated and validated circuit behaviour in LTspice before schematic capture and PCB layout, preparing the design for physical prototyping.
- Designed a custom battery-temperature monitoring PCB in KiCad, including the schematic and board layout
- Incorporated CAN and UART interfaces for communication between the temperature-monitoring system and the Battery Management System.

Poker Decision Simulator

Java, JUnit, Monte Carlo Simulation

- Developed a Texas Hold'em simulator with complete betting rounds, hand evaluation and AI-controlled opponents.
- Implemented Monte Carlo simulation to estimate hand equity and support decisions under incomplete information.
- Built an explainable fold/call/raise engine using pot odds, expected value, object-oriented design and JUnit tests.

Delivery Robot & Navigation Platform

In Progress

Java, Spring Boot, PostgreSQL, ESP32, GPS

- Developing an ESP32-based robot with GPS positioning, motor control, obstacle detection and wireless telemetry.
- Building a Spring Boot REST API for robot control, route planning, delivery management and live status monitoring.
- Designing a PostgreSQL data model for robots, deliveries, GPS waypoints, telemetry and route history, with ESP32-backend integration.

Interactive Pixel-Art Portfolio Website

Frontend Development

- Designed and developed a responsive pixel-art portfolio with a retro desktop UI, custom sprite animations and cross-browser compatibility for Safari and Firefox.

ACHIEVEMENTS & INTERESTS

Awards: British Physics Olympiad (BPhO) - Gold; Gold CREST Award for QinetiQ project.

Scholarships: University of Warwick Global Excellence Scholarship; Malvern College Honorary Scholarship.

Sponsorship: Secured approximately £1,000 from KT Lab Cloud and Chevron at the Enactus National Expo Kazakhstan.

Music: Graduated from music school in Kazakhstan (piano), with a final average of 5.0/5.0.

Activities & interests: Choir and orchestra, Malvern College; volleyball, University of Warwick; robotics.